

SEMESTER END EXAMINATIONS – FEBRUARY 2024

Program	: B.E. - Common to all Program	Semester	: V
Course Name	: Research Methodology and Intellectual Property Rights	Max. Marks	: 100
Course Code	: AL58	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Suitably assume any missing data.
- Use of statistical tables are permitted.

UNIT - I

- a) What do you mean by research misconduct? Briefly explain different types of research misconduct. CO1 (08)
 - b) Distinguish between the following: CO1 (12)
 - i) Descriptive VS Analytical research
 - ii) Applied VS Fundamental research
 - iii) Quantitative VS Qualitative research
 - iv) Conceptual VS Empirical research.
- a) What do you mean by bibliographic database? Explain different types of bibliographic database. CO1 (10)
 - b) Explain in detail, knowledge flow through citation. CO1 (10)

UNIT - II

- a) Write short note on: CO2 (10)
 - i) Functions of hypothesis
 - ii) Types of hypotheses.
 - b) What is factorial design. Explain its advantages and disadvantages. CO2 (10)
- a) Explain the meaning of the following in context of research: CO2 (06)
 - i) Extraneous variable
 - ii) Experimental and control group.
 - b) Compare and contrast random block design and Latin square design with its merits and demerits. CO2 (14)

UNIT - III

5. a) Explain any two methods of collecting primary data in research. CO3 (08)
b) Explain the need for sampling. Enumerate the characteristics of a good sample design. CO3 (06)
c) Explain the nominal and ordinal scales of measurement. Provide suitable examples for each. CO3 (06)
6. a) The mean of a certain production process is known to be 50 with a standard deviation of 2.5. The production manager may welcome any change in mean value towards higher side but would like to safeguard against decreasing values of mean. He takes a sample of 12 items that gives a mean value of 48.5. What inference should the manager take for the production process on the basis of sample results? Use 5 percent level of significance for the purpose. CO3 (10)
b) Write a short note on ANOVA and ANOCOVA, highlighting their uses in research. CO3 (10)

UNIT- IV

7. a) Explain the various types of Intellectual Property Rights. CO4 (06)
b) Define Intellectual Property Rights. Discuss the need for intellectual property laws in a developing country. CO4 (04)
c) Explain Basmati Rice patent case in detail. CO4 (10)
8. a) Explain the different types of patent applications in India. CO4 (10)
b) Discuss the role of Patent Cooperation Treaty in the acquisition of international patent rights. CO4 (10)

UNIT- V

9. a) What are the essential requirements of a registrable design? Explain. CO5 (10)
b) What is a trademark? How are trademarks protected? CO5 (05)
c) What is an infringement of trademark? Explain with an example. CO5 (05)
10. a) What is a copyright? What are its characteristics? CO5 (10)
b) What is infringement of copyrights? What are its remedies? CO5 (10)
